The Strong CP Problem Is Not a Problem

θ_QCD = 0 as the Ground State of an Integer-Topological System

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I. ABSTRACT

The strong CP problem asks: why is the QCD vacuum angle θ so close to zero? This paper argues that the question contains a methodological error and that the "problem" dissolves under examination of its own premises.

Every measurement is consistent with θ = 0 exactly. The topological structure of QCD has sectors labeled by integers. The energy of the vacuum as a function of θ is $E(\theta) = E_0 - \chi\_top \cdot \cos(\theta)$, minimized at θ = 0. The vacuum is defined as the minimum energy state. Therefore the vacuum has θ = 0.

This paper does not claim that θ is a dynamical variable that evolves toward zero. That is the Peccei-Quinn mechanism, which promotes θ to a field. This paper claims that the vacuum is identified with the energy minimum — the same identification performed for the Higgs vacuum at v = 246 GeV, for the QED vacuum, and for every other vacuum state in quantum field theory. The institution performs this identification without comment in every other sector of the Standard Model. The institution refuses to perform it for θ and declares a problem instead. The burden of proof is on those who claim the universe is not in its ground state, not on those who observe that it is.

The institution's declaration of a problem rests on six pillars, each examined and found to depend on the same error: treating the model's freedom to write θ ≠ 0 as the universe's freedom to realize θ ≠ 0. The Lagrangian permits θ ∈ [0, 2 π) — but the Lagrangian also permits α ≠ 1/137, and no mechanism is demanded for that far more mysterious value. The institution objects that θ = 0 is a symmetry-enhanced point unlike 1/137 — but CP symmetry protecting θ = 0 is the explanation, not the mystery. The decomposition θ_phys = θ_bare + arg(det M_q) involves two "independent" quantities that cancel — but only θ_phys is observable, and the individual pieces are gauge-dependent, changing

under chiral field redefinitions. Naturalness declares $\theta = 0$ unnatural — but naturalness has produced zero confirmed predictions in forty years. The θ-vacuum formalism permits any value — but the formalism is a map, and maps showing roads not taken do not require explanation. The Peccei-Quinn mechanism elegantly solves the problem — but an unfound solution to a non-problem is not evidence that the problem exists.

The strong CP problem dissolves when the distinction between model freedom and physical freedom is maintained, and when the identification vacuum = ground state is applied consistently across the Standard Model rather than selectively withheld from one parameter. $\theta_{QCD} = 0$ is the first of the 19 Standard Model parameters addressed in this series. It is an integer. It is the ground state. It is the most fundamental value a parameter can take. The institution accepts the unexplained value 1/137.035999 without declaring a problem. The institution declares a fifty-year problem for the value 0. The priorities are inverted.

II. WHAT IS MEASURED

The QCD vacuum angle θ enters physics through one observable: the neutron electric dipole moment. If $\theta \neq 0$, the neutron acquires an electric dipole moment proportional to θ:

$d_n \approx \theta \times 3.6 \times 10^{-16}$ e·cm

The current experimental bound is $|d_n| < 1.8 \times 10^{-26}$ e·cm (Abel et al., 2020). This implies:

$|\theta| < 5 \times 10^{-11}$ Every measurement of every observable sensitive to θ is consistent with $\theta = 0$ exactly. No experiment has ever produced evidence for $\theta \neq 0$. No observation at any energy scale, in any process, at any precision, has required a nonzero θ to explain. The experimental situation is unambiguous: $\theta = 0$ to the limit of measurement.

III. WHAT IS DECLARED

The institution declares the strong CP problem: why is θ so small?

The problem was formulated by 't Hooft (1976), who showed that instanton effects make the θ-vacuum physically meaningful, and sharpened by Peccei and Quinn (1977), who proposed a dynamical solution. The problem has been a central question in particle physics for nearly fifty years. It has generated thousands of papers, multiple proposed solutions (the axion, Nelson-Barr models, spontaneous CP violation), and dedicated experimental programs (axion searches at ADMX, CASPEr, ABRACADABRA, and others).

The problem is stated as follows. The QCD Lagrangian contains a term:

$L_\theta = (\theta/32\pi^2) \cdot G^a{}_{\mu\nu} \cdot \tilde{G}^{a,\mu\nu}$

where G is the gluon field strength tensor and $\tilde{G}$ is its dual. This term is gauge-invariant, Lorentz-invariant, and renormalizable. Nothing in the Standard Model forbids it. The coefficient θ is a free parameter. It could take any value in $[0, 2\pi)$. It takes the value 0 (or extremely close to 0). The question is why.

This paper examines the logical structure of this declaration and finds that the "problem" is generated by a specific assumption that is not supported by observation.

IV. THE IDENTIFICATION THAT THE INSTITUTION PERFORMS EVERYWHERE ELSE


Vacuum = Minimum Energy State: Applied Everywhere Except θ

Sector	Energy functional	Minimum	Vacuum = minimum?	Problem declared?
Higgs	$V(\varphi) = \mu^2\lvert\varphi\rvert^2 + \lambda\lvert\varphi\rvert^4$	v = 246 GeV	YES	NO
QED	H_QED ground state	No real particles	YES	NO
QCD (perturbative)	Λ_QCD ground state	Confined vacuum	YES	NO
QCD (topological)	$E(\theta) = E_0 - \chi\cos(\theta)$	θ = 0	NO — refused!	YES — for 50 years!

The same identification is performed without comment in every row except the last. The refusal to apply it IS the "problem."

Figure 1: Fig. 7: Higgs, QED, and perturbative QCD vacua are identified with their energy minima without comment. Only the θ-vacuum is refused this identification — and the refusal is called a "problem."

Before examining the assumption that generates the problem, we establish the operation the institution performs for every other vacuum in quantum field theory and then refuses to perform for θ.

4.1 The Higgs Vacuum

The Higgs potential is $V(\varphi) = \mu^2|\varphi|^2 + \lambda|\varphi|^4$ with $\mu^2 < 0$. The minimum is at $|\varphi| = v = \sqrt{(-\mu^2/2\lambda)} = 246$ GeV. The institution identifies the vacuum with this minimum. No mechanism is demanded for how the Higgs field "reached" v = 246 GeV. No one asks why the field "chose" this value rather than sitting at φ = 0 or φ = 500 GeV. The identification is: the vacuum is the minimum energy state, the minimum is at v = 246 GeV, the vacuum has v = 246 GeV. Done.

The Higgs field is a dynamical field with an equation of motion. But the identification of the vacuum with the minimum does not invoke that equation of motion. The identification is definitional: vacuum means minimum energy. The dynamics are relevant to fluctuations around the vacuum, not to the identification of the vacuum itself.

4.2 The QED Vacuum

The QED vacuum is the state with no real photons and no real electron-positron pairs. The vacuum energy includes zero-point fluctuations and virtual pair contributions. The vacuum is the minimum energy state of the QED Hamiltonian. No mechanism is demanded for why the QED vacuum is at its minimum rather than at some excited state with a background electric field.

4.3 The QCD Vacuum at θ = 0

The QCD vacuum energy as a function of θ is $E(\theta) = E_0 - \chi\_top \cdot \cos(\theta)$. The topological susceptibility χ_top is positive (confirmed by lattice QCD and the Witten-Veneziano relation for the η' mass). The minimum is at θ = 0. The identification is: the vacuum is the minimum energy state, the minimum is at θ = 0, the vacuum has θ = 0.

The operation is identical to the Higgs case. The energy functional is known. The minimum is computed. The vacuum is identified with the minimum. The identification is definitional.

4.4 The Double Standard

The institution performs this identification for the Higgs vacuum without comment. The institution refuses to perform it for the θ-vacuum and declares a problem instead.

The institution may respond that θ is a parameter, not a field, and therefore the "minimum of E(θ)" is not the same operation as "minimum of V(φ)." This response confuses the representation with the physics. E(θ) is the energy of the vacuum labeled by θ. V(φ) is the energy of the field configuration φ. In both cases, the vacuum is the configuration of lowest energy. Whether the variable being minimized is called a "field" or a "parameter" does not change the physical content of the identification: the vacuum is the minimum.

If the institution insists that a dynamical mechanism is required for the vacuum to "reach" θ = 0, the same demand applies to the Higgs field reaching v = 246 GeV. If no mechanism is required for the Higgs, no mechanism is required for θ. The standards must be consistent.

4.5 The Superselection Question

The θ-vacua are sometimes described as superselection sectors — states between which no transition is possible. If so, the question becomes: which sector is the universe in?

The answer is: the lowest energy sector. This is the definition of the vacuum in quantum field theory. If the universe were in the $\theta = 0.5$ sector, that sector would have energy $E_0 - \chi\_top \cdot \cos(0.5) > E_0 - \chi\_top$, which is higher than the $\theta = 0$ sector. The vacuum is the lowest energy state. The lowest energy state is $\theta = 0$.

Asking why the universe is in the $\theta = 0$ sector is asking why the universe is in the vacuum. The universe is in the vacuum because that is the ground state. Asking why the universe is in the ground state rather than an excited state is a question with a universal answer: because systems are in their ground states unless something has excited them. Nothing in the Standard Model excites the QCD vacuum out of the $\theta = 0$ sector. If someone wishes to claim the universe is in an excited θ-sector, the burden is on them to provide the excitation mechanism.

The burden of proof is inverted from how the institution frames it. The institution demands a mechanism to explain the ground state. Physics demands a mechanism to explain departures from the ground state.

V. THE ASSUMPTION THAT GENERATES THE PROBLEM

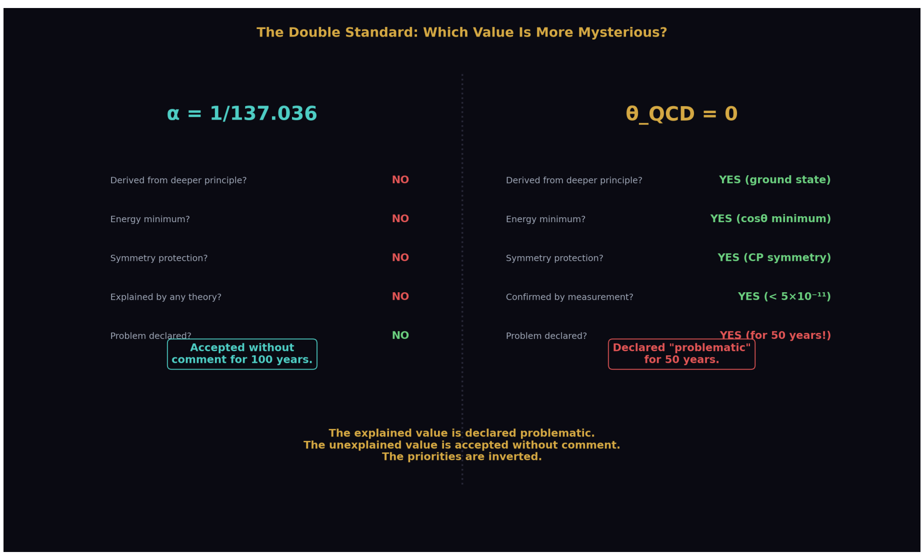

Figure 2: Fig. 6: α=1/137 has no derivation, no symmetry protection, no energy minimum — accepted without comment. θ=0 has all three — declared “problematic” for 50 years.

The assumption is: because the model parameter θ can take any value in $[0, 2\pi)$, the physical quantity θ could take any value in $[0, 2\pi)$, and the fact that it takes the value 0 requires a mechanism or explanation.

This assumption treats model freedom as physical freedom. The two are not the same.

A model is a mathematical structure that reproduces observations. The parameters of the model are coefficients in the mathematical structure. The values of those parameters are determined by measurement. The model permits a range of values. The universe realizes one value. The gap between “the model permits” and “the universe realizes” is not a problem. It is the definition of a free parameter. Every free parameter in the Standard Model has this gap. The model permits any value. The universe picks one. The physicist measures it.

The institution does not apply the “problem” declaration consistently.

The fine structure constant $\alpha = 1/137.035999$ is a free parameter. The Lagrangian permits any positive value. The value 1/137.035999 has no known derivation from deeper principles. No mechanism has been proposed to explain why α takes this value rather than 1/100 or 1/200. The institution does not declare an “α problem.” The institution accepts that α is measured and moves on.

The institution may respond that $\theta = 0$ is different because it is a symmetry-enhanced point — the CP-conserving value — while $\alpha = 1/137$ is a generic point with no symmetry enhancement. This observation is correct. But it supports the argument rather than undermining it. A symmetry-enhanced point is a point with additional structure — additional reason to be there. CP symmetry protects $\theta = 0$. The energy minimum is at $\theta = 0$. The symmetry and the energy minimum coincide. The parameter with the most structural reasons to take its value is the one the institution declares most problematic. The parameter with no structural reason at all — $\alpha = 1/137.035999$ — is accepted without comment.

The electron mass $m_e = 0.511$ MeV is a free parameter determined by the electron Yukawa coupling. The Lagrangian permits any value. No mechanism is demanded. The top quark mass $m_t = 172.69$ GeV is a free parameter. No mechanism is demanded. None of the 18 other Standard Model parameters triggers a “problem” declaration by virtue of being a free parameter. Only θ, the one parameter with the most explicable value — zero, the ground state, the energy minimum, the CP-conserving point — is declared problematic.

VI. THE GROUND STATE ARGUMENT

6.1 Integer Topological Sectors

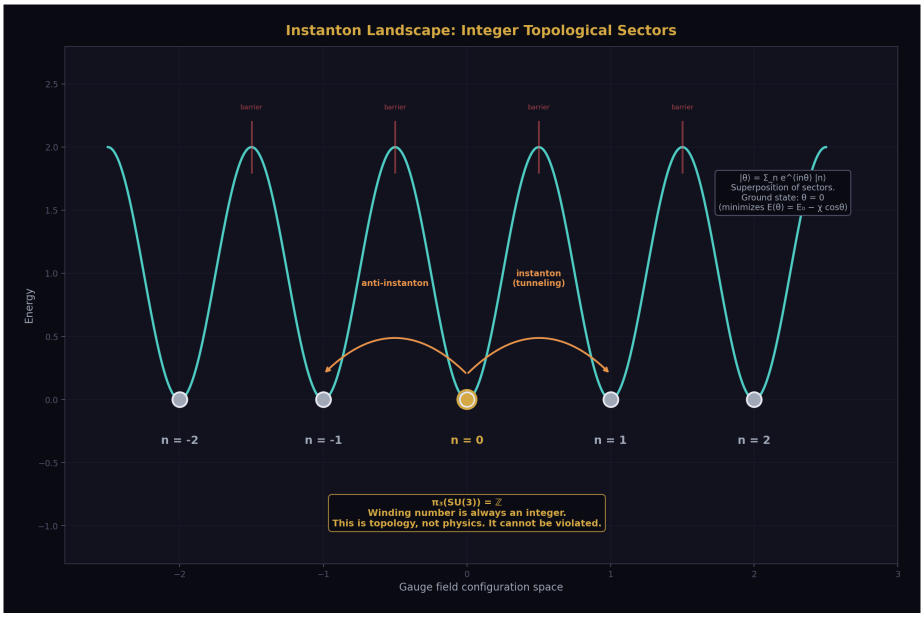

Figure 3: Fig. 3: Periodic energy barriers between integer topological sectors n$\in \mathbb{Z}$. Instantons tunnel between adjacent sectors. The ground state at n=0 is the minimum of the cosine potential.

The QCD vacuum has a topological structure established by the institution's own mathematics and confirmed by lattice QCD calculations.

The gauge field configurations of SU(3) fall into topological sectors labeled by an integer n — the winding number, also called the Pontryagin index or topological charge. This integer counts the number of times the gauge field wraps around the group manifold. It is an integer by mathematical theorem: the third homotopy group π_3(SU(3)) = $\mathbb{Z}$. The winding number of a continuous map from S^3 to SU(3) is always an integer. This is topology, not physics. It cannot be violated.

The sectors are separated by energy barriers. Transitions between sectors occur through instantons — localized gauge field configurations that carry one unit of topological charge. Instantons have been studied extensively in the QCD literature and confirmed in lattice calculations.

6.2 The Energy Functional

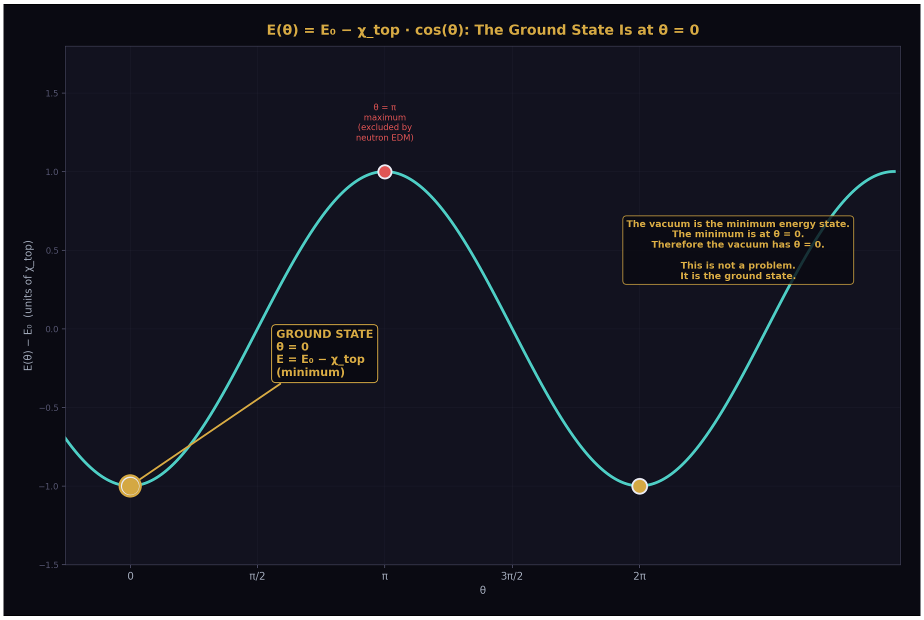

Figure 4: Fig. 1: The cosine energy functional with its unique minimum at θ=0. The vacuum is the minimum energy state. Therefore the vacuum has θ=0. This is not a problem — it is the ground state.

The energy of the QCD vacuum as a function of θ is:

$E(\theta) = E_0 - \chi\_top \cdot \cos(\theta)$

where χ_top is the topological susceptibility — a positive quantity computed on the lattice and measured indirectly through the η' meson mass via the Witten-Veneziano relation. The positivity of χ_top is established.

The function $-\cos(\theta)$ has a unique global minimum on $[0, 2\pi)$ at $\theta = 0$.

6.3 The Identification

The physical vacuum is the state of minimum energy. This is the definition of the vacuum in quantum field theory — the same definition used for the Higgs vacuum, the QED vacuum, and every other vacuum state. The energy is minimized at $\theta = 0$. Therefore the vacuum has $\theta = 0$.

This is not a dynamical claim. The paper does not assert that θ was once nonzero and evolved toward zero. That is the Peccei-Quinn mechanism, which promotes θ to a dynamical field (the axion) and provides explicit relaxation dynamics. This paper makes a different and simpler claim: the vacuum is defined as the minimum energy state, the minimum is at θ = 0, therefore the vacuum has θ = 0. The identification is the same operation the institution performs for every other vacuum in the Standard Model without demanding a dynamical narrative.

The ground state requires no explanation beyond the energy functional. The energy functional is known. The minimum is at θ = 0. The system is at the minimum. This is the complete account.

6.4 The Analog

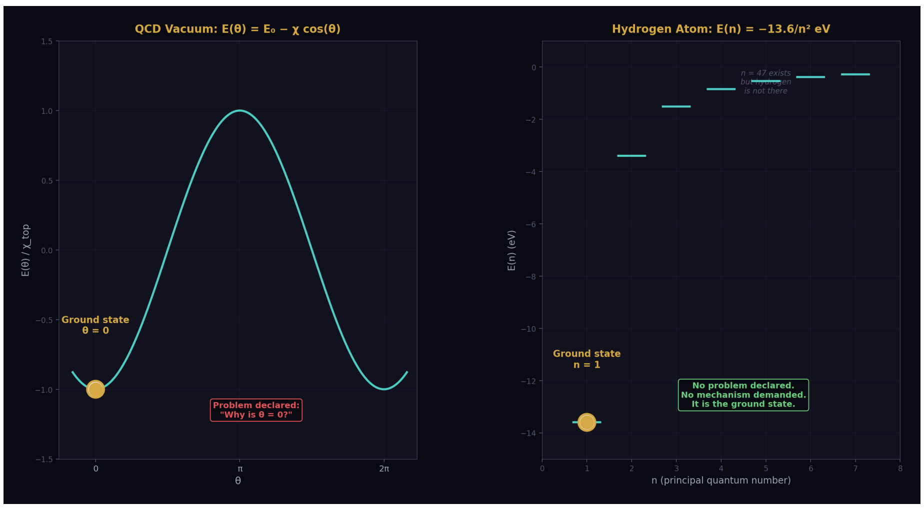

Figure 5: Fig. 2: E(θ) with minimum at 0 (left) and hydrogen E(n) with minimum at n=1 (right). Both are ground states. Neither requires a mechanism. Only one has a "problem" declared.

The hydrogen atom has states $|n\rangle$ for every positive integer n. The ground state is $|n = 1\rangle$. The institution does not declare a "hydrogen ground state problem." The institution does not ask why hydrogen "chose" n = 1 rather than n = 47. The institution does not demand a mechanism for how hydrogen "reached" n = 1. The ground state is the ground state. The energy is lowest there. The question "why n = 1?" is answered by "because that minimizes the energy."

The point of the analogy is not about relaxation dynamics. Hydrogen can reach n = 1 by radiating photons, but that is irrelevant. Whether hydrogen was prepared in n = 1 or arrived there by radiation, the ground state does not require explanation. No one demands a mechanism to explain a ground state in any other system. The logical status of θ = 0 as the QCD ground state is identical to the logical status of n = 1 as the hydrogen ground state. Both are energy minima. Neither requires explanation beyond the energy functional.

VII. THE BASIS DECOMPOSITION OBJECTION

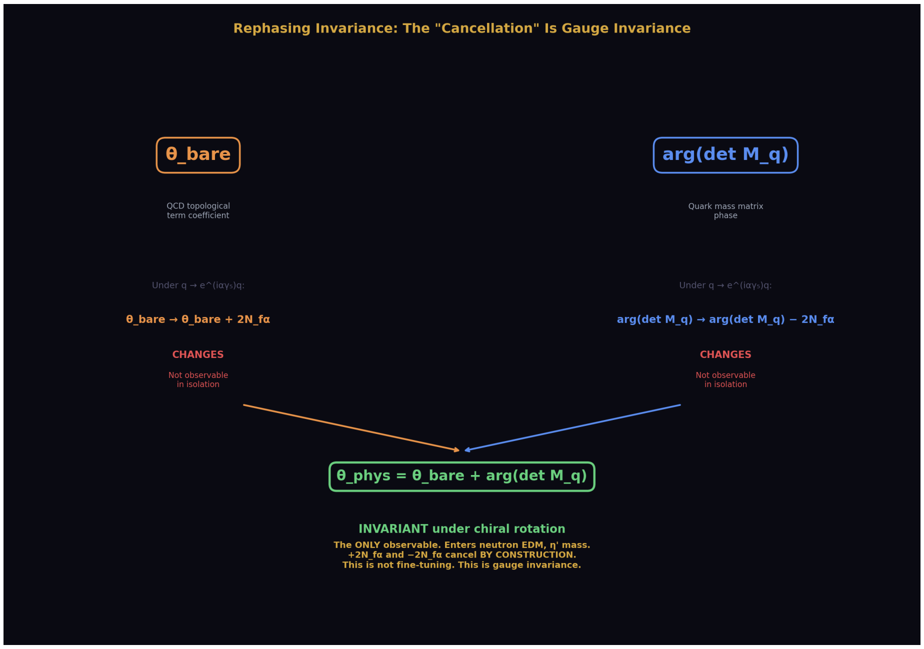

Figure 6: Fig. 4: θ_bare and arg(det M_q) transform oppositely under chiral rotation — their sum θ_phys is invariant. The "fine-tuning" is gauge artifacts canceling by construction.

7.1 The Objection

The institution's primary response to the ground state argument is the basis decomposition. The physical θ is a sum:

θ_phys = θ_bare + arg(det M_q)

where θ_bare is the coefficient of the topological term in the QCD Lagrangian and arg(det M_q) is the phase of the determinant of the quark mass matrix, which comes from the Yukawa sector. The institution argues: these come from independent sectors of the Standard Model, there is no reason for them to cancel, yet they cancel to 10^{-10} precision. This cancellation is the fine-tuning.

7.2 The Two-Step Response

The response has two distinct steps that must not be conflated.

Step 1: The fine-tuning framing is wrong. The individual pieces θ_bare and arg(det M_q) are not independently physical. Under a chiral rotation of the quark fields q → e^(iαγ_5)q, the two components transform:

θ_bare → θ_bare + 2N_f α

arg(det M_q) → arg(det M_q) − 2N_f α

The sum θ_phys is invariant. The individual pieces are not — they depend on the choice of field basis. A quantity that changes under a field redefinition is not a physical observable. It is a gauge artifact. The "cancellation" between θ_bare and arg(det M_q) is a cancellation between gauge-dependent quantities. Gauge-dependent quantities cancel by construction whenever they appear in a gauge-invariant combination. That is what gauge invariance means. The fine-tuning framing collapses because you cannot fine-tune gauge artifacts.

Step 2: The remaining question is answered by the ground state. Step 1 kills the fine-tuning version of the problem — the version that says "two independent quantities cancel." It does not by itself explain why θ_phys = 0. The gauge-invariant sum θ_phys could be gauge-invariantly equal to 3.7. The question "why is θ_phys = 0?" survives Step 1. It is answered by Section VI: the vacuum is the minimum energy state, and the energy minimum is at θ_phys = 0.

These two steps are distinct. Step 1 addresses the fine-tuning framing. Step 2 addresses the value. Conflating them — demanding that the rephasing argument alone explain why $\theta = 0$ — misidentifies what each step does.

7.3 The Observability Test

Can the institution exhibit a physical observable that depends on θ_bare alone, independent of arg(det M_q)?

No. Every CP-violating observable in QCD depends on θ_phys = θ_bare + arg(det M_q). The neutron electric dipole moment depends on θ_phys. The η' mass depends on θ_phys through the topological susceptibility. No process, no measurement, no observable separates the two contributions. They are not independently measurable because they are not independently physical. The "independence" of the two sectors is a statement about the model's organizational structure, not about the physics.

VIII. THE NATURALNESS OBJECTION

8.1 The Objection

The naturalness principle states: a parameter being zero or extremely small without a symmetry reason is "unnatural" and requires explanation. Since no obvious symmetry of the Standard Model forces $\theta = 0$, the smallness of θ is unnatural.

8.2 The Track Record

Naturalness as a predictive principle has been tested by experiment.

Naturalness predicted superpartners at or below the TeV scale to stabilize the Higgs mass. None have been found through two full LHC runs.

Naturalness predicted new physics at the TeV scale to address the hierarchy problem. No new physics has been found at the LHC beyond the Standard Model Higgs.

The cosmological constant violates naturalness by 120 orders of magnitude. No naturalness-based resolution has been confirmed.

The track record over forty years is zero confirmed predictions. Naturalness is a heuristic, not a law. When it contradicts measurement, measurement wins. The measurement says $\theta = 0$.

8.3 The Symmetry That Naturalness Says Is Absent

The naturalness objection states that no symmetry forces $\theta = 0$. This is incorrect.

CP symmetry forces $\theta = 0$. If CP is an exact symmetry of the strong interaction, the topological term $(\theta/32\pi^2)\, G\cdot\tilde{G}$ is forbidden because $G\cdot\tilde{G}$ is CP-odd. The only CP-invariant value of θ is 0 (or π, which corresponds to spontaneous CP violation in the strong sector and is excluded by the neutron EDM measurement).

The data says: CP is conserved in the strong sector to 10^{-10} precision. The institution knows this. It is the basis of the Nelson-Barr approach. The simplest interpretation is that CP is an exact symmetry of QCD and $\theta = 0$ is protected by it.

The institution objects: CP is violated in the weak sector through the CKM phase. How can CP be exact in the strong sector?

The answer: strong CP violation and weak CP violation are different observables. The CKM phase produces CP violation in weak processes — K meson mixing, B meson decays. It does not produce CP violation in strong processes — no neutron EDM is observed. The data shows exact strong CP conservation coexisting with weak CP violation. These are not contradictory. They are different measurements of different quantities in different sectors, and the measurements are consistent with each other and with the Standard Model.

The CKM phase enters arg(det M_q) through the Yukawa couplings. But as established in Section VII, arg(det M_q) is gauge-dependent and unobservable in isolation. Only θ_phys enters observables. θ_phys = 0. The weak CP violation does not "feed into" strong CP violation — it feeds into one gauge-dependent piece of θ_phys, and the physical combination remains zero. This is not a cancellation. It is gauge invariance operating as it always does.

IX. THE θ-VACUUM OBJECTION

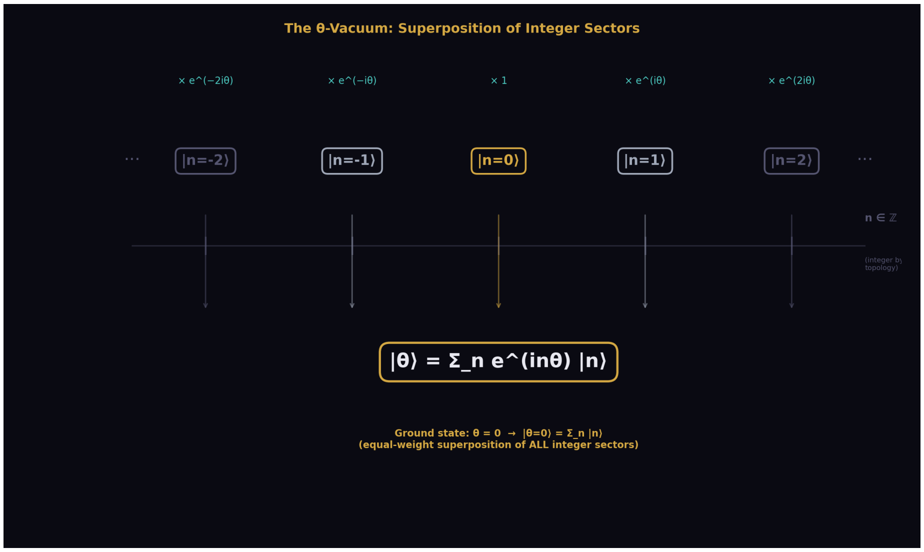

Figure 7: Fig. 5

9.1 The Objection

The θ-vacuum is a superposition of topological sectors:

$|\theta\rangle = \Sigma$ n e^(inθ) $|n\rangle$ This is a valid quantum state for any $\theta \in [0, 2\pi)$. The existence of these states means the vacuum "could be" at any θ. Why is it at θ = 0?

9.2 The Response

The hydrogen atom has states $|n\rangle$ for every positive integer n. The state $|n = 47\rangle$ is a valid quantum state. The existence of $|n = 47\rangle$ does not mean hydrogen should be in $|n = 47\rangle$. Hydrogen is in $|n = 1\rangle$ because that is the ground state.

The θ-vacuum $|\theta\rangle$ exists for every θ. The ground state is $|\theta = 0\rangle$ because $E(\theta) = E_0 - \chi$_top cos(θ) is minimized at θ = 0. The existence of $|\theta = \pi/7\rangle$ does not mean the vacuum should be at $\theta = \pi/7$ any more than the existence of $|n = 47\rangle$ means hydrogen should be at n = 47.

The formalism permits the mathematical construction of excited states. The physics selects the ground state. The formalism is a map that includes all possible states. The physics is the territory where the system occupies one state — the state of lowest energy. Maps showing roads not taken do not require explanation.

X. THE AXION OBJECTION

10.1 The Objection

The Peccei-Quinn mechanism elegantly solves the strong CP problem by introducing a U(1)_PQ symmetry whose spontaneous breaking produces the axion — a light pseudoscalar that dynamically relaxes θ to zero. The elegance of the solution suggests the problem is real.

10.2 The Response

The logical structure of this argument is: an elegant solution exists, therefore the problem exists. This is circular. The existence of a solution does not validate the problem.

The axion has been searched for experimentally for over forty years. ADMX, HAYSTAC, CASPEr, ABRACADABRA, CAST, IAXO, and numerous other experiments have searched for axions across a wide range of parameter space. No axion has been detected. The absence of detection after forty years of dedicated searching means the axion cannot be cited as evidence that the strong CP problem is real. An unfound particle predicted by an unconfirmed problem is not evidence for the problem.

10.3 Compatibility

If the axion is discovered and its couplings match the Peccei-Quinn prediction, the PQ mechanism is confirmed. In that case, both the PQ mechanism and the ground state argument are valid simultaneously — the PQ mechanism provides dynamics for how the vacuum reaches the ground state, and the ground state argument explains why the ground state is at $\theta = 0$. The two are compatible, not competing.

This paper argues that the ground state argument is sufficient without the PQ mechanism. It does not argue that the PQ mechanism is wrong. The axion may exist for reasons independent of the strong CP problem — it is a viable dark matter candidate and appears in string compactifications. Its existence is an empirical question that should be investigated on its own merits.

XI. THE SOCIOLOGY OBJECTION

The response "if this were so simple, someone would have said it already" is not a physics argument. It is a sociological argument. Sociological arguments do not determine the truth of physical claims.

Versions of the ground state argument have appeared in the literature. The observation that $\theta = 0$ is the energy minimum is stated in every QFT textbook that derives the θ-vacuum. The observation that the basis decomposition involves gauge-dependent quantities is stated wherever the rephasing invariance of θ_phys is demonstrated. What has not been stated with the necessary directness is: these observations, taken together, dissolve the problem rather than solve it. The institution has treated each observation as a piece of background rather than as the complete answer.

The ground state is the complete answer. This paper states it as such.

XII. THE INTEGER CONTENT

θ_QCD = 0 is the first of the 19 Standard Model parameters addressed in this series.

The topological sectors are labeled by integers: $n \in \mathbb{Z}$. The winding number is an integer by mathematical theorem. The ground state is at the integer 0 — the additive identity, the trivial element of ($\mathbb{Z}$, +).

In the framework of [HOWL-PHYS-1-2026] through [HOWL-PHYS-6-2026], the laws are integers and the universe supplies measured values. For θ_QCD, the law (integer topological sectors, cosine energy functional) and the value (0) are both integers. There is no gap between the law and the parameter. This is the simplest possible case: the ground state of an integer system is the integer zero.

XIII. WHAT THIS PAPER DOES NOT CLAIM

This paper does not claim the axion does not exist. The axion is a particle whose existence is an empirical question independent of the strong CP problem.

This paper does not claim the Peccei-Quinn mechanism is wrong. If the axion exists, the PQ mechanism may correctly describe dynamics that supplement the ground state identification. Both may be true simultaneously.

This paper does not claim the Nelson-Barr mechanism is wrong. Nelson-Barr provides specific UV symmetry structure compatible with $\theta = 0$.

This paper does not claim any new physics. Every equation is from the institution's own literature. The topological sectors, the energy functional, the rephasing invariance, the lattice calculations — all standard results. The paper reorganizes existing findings and draws a conclusion the institution has not drawn: the strong CP problem is generated by a methodological error, not by the physics.

XIV. FALSIFICATION CRITERIA

F1. If θ_phys ≠ 0 is measured — if a neutron electric dipole moment is detected at a value implying $|\theta| > 10^{-10}$ — the ground state argument is falsified. θ_phys ≠ 0 means the vacuum is not at the energy minimum, which requires either a different energy functional or excitation out of the ground state. Either finding requires new physics.

F2. If a physical observable is identified that depends on θ_bare alone, independent of arg(det M_q), the basis decomposition objection is rehabilitated. The two pieces would be independently physical, and their cancellation would require explanation. No such observable is currently known.

F3. If naturalness produces a confirmed prediction in the strong CP sector — if a particle or effect predicted by naturalness arguments specifically applied to θ is observed — the naturalness critique is weakened for this domain.

F4. If the axion is discovered with couplings matching the PQ prediction, the PQ mechanism is confirmed as additional correct physics. This does not falsify the ground state argument but establishes that additional dynamics exist beyond the ground state identification.

F5. If the topological susceptibility χ_top is measured or computed to be negative, the energy minimum shifts from θ = 0 to θ = π , and the ground state argument predicts θ = π , which is falsified by the neutron EDM measurement. Current lattice calculations confirm χ_top > 0.

XV. CONCLUSION

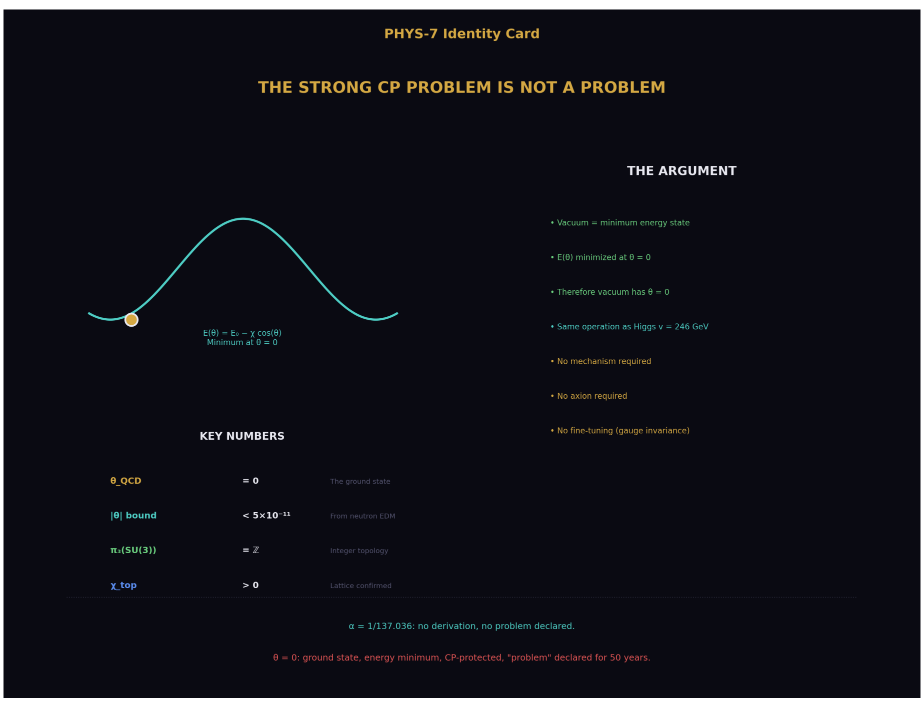

Figure 8: Fig. 8: θ=0 is the ground state of an integer-topological system — the energy functional, the integer topology, the double standard, and no mechanism required.

The strong CP problem asks why θ_QCD = 0. The answer is: because 0 is the ground state.

The vacuum is the minimum energy state. The energy $E(\theta) = E_0 - \chi\_top \cdot \cos(\theta)$ is minimized at θ = 0. The vacuum has θ = 0. This is the same identification the institution performs for the Higgs vacuum at v = 246 GeV — without comment, without a declared problem, without demanding a mechanism. The only difference is that the institution has spent fifty years refusing to apply to θ the same operation it applies everywhere else.

The fine-tuning version of the problem — that θ_bare and arg(det M_q) cancel — dissolves because the individual pieces are gauge-dependent. They are not independently physical. Their cancellation is guaranteed by gauge invariance, not by conspiracy.

The naturalness version of the problem — that θ = 0 is "unnatural" — fails because naturalness has zero confirmed predictions, and because CP symmetry does protect θ = 0, contrary to the claim that no symmetry is present.

The institution accepts α = 1/137.035999 — an unexplained, underived number — without declaring a problem. The institution declares a fifty-year problem for the value 0 — the additive identity, the energy minimum, the CP-conserving point, the ground state. The explained value is declared problematic. The unexplained value is accepted without comment. If any value in the Standard Model warrants investigation, it is 1/137.035999, not 0.

θ_QCD = 0. It is an integer. It is the ground state. It is the minimum of the energy functional. It is protected by CP symmetry. It is confirmed by every measurement. It requires no mechanism, no axion, no new symmetry, and no fine-tuning. It requires only the recognition that the universe is in its ground state, and that being in the ground state is not a problem.

APPENDIX A: THE DOUBLE STANDARD

Parameter	Measured value	Model permits	Derived from deeper principle?	Problem declared?	Why or why not?
α EM	1/137.036	Any positive value	No	No	Accepted as measured
m e	0.511 MeV	Any positive value	No	No	Accepted as measured
m t	172.69 GeV	Any positive value	No	No	Accepted as measured

Parameter	Measured value	Model permits	Derived from deeper principle?	Problem declared?	Why or why not?
$\sin^2\theta$ W	0.2312	Any value in (0,1)	Partially (3/8 at GUT scale)	No	Accepted as measured
v	246 GeV	Any positive value	No	Yes (hierarchy problem)	Radiative instability, not the free parameter itself
θ QCD	0	[0, 2 π)	Yes (ground state of energy functional)	Yes (strong CP problem)	The only parameter with a derivation is the only one declared problematic

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APPENDIX B: THE GROUND STATE COMPARISON

System	Ground state	Excited states exist?	Mechanism to reach ground state required?	Problem declared?
Hydrogen atom	n = 1	Yes	No (it's the ground state)	No
Harmonic oscillator	n = 0, x = 0	Yes	No	No
Crystal lattice	Lowest configuration	Yes (defects, phonons)	No	No
Higgs vacuum	v = 246 GeV	Yes (φ = 0)	No (vacuum = minimum of V(φ))	No (hierarchy problem is about radiative stability, not about the identification)
QED vacuum	No real particles	Yes (any particle state)	No	No

System	Ground state	Excited states exist?	Mechanism to reach ground state required?	Problem declared?
QCD vacuum	$\theta = 0$	Yes ($\theta \in (0, 2\pi)$)	No — but the institution demands one	Yes

In every other system, the ground state identification is accepted. Only for the QCD vacuum is the ground state declared a problem requiring a mechanism.

APPENDIX C: THE NATURALNESS SCORECARD

Naturalness prediction	Year	Expected observation	Actual observation	Status
Superpartners at TeV scale	1980s	SUSY particles at LEP/Tevatron/LHC	None found through LHC Run 2	Failed
New physics at TeV scale	1990s	BSM particles or effects at LHC	None found	Failed
Large θ QCD	1976	$\theta \sim O(1)$ unless mechanism intervenes	$\theta < 10^{-10}$	"Problem" declared rather than prediction acknowledged as failed
Cosmological constant ~ M_{Pl}^4	1980s	$\Lambda \sim (10^{18}\ \mathrm{GeV})^4$	$\Lambda \sim (10^{-3}\ \mathrm{eV})^4$, 120 orders smaller	Failed
Light Higgs requires new physics	2000s	m_H stabilized by new particles	$m_H = 125$ GeV, no stabilizing particles found	Failed

Zero confirmed predictions in forty years.

APPENDIX D: THE REPHASING INVARIANCE

Under a chiral field rotation $q_f \to e^{i\alpha_f \gamma_5} q_f$:

Quantity	Transformation	Observable?
θ bare	θ bare $\rightarrow$ θ bare + 2α f	No — changes under field redefinition
arg(det M q)	arg(det M q) $\rightarrow$ arg(det M q) $-$ 2α f	No — changes under field redefinition
θ phys = θ bare + arg(det M q)	Invariant	Yes — enters all observables
d n (neutron EDM)	Depends on θ phys only	Yes
η’ mass	Depends on χ top, related to θ phys = 0	Yes

The individual pieces are gauge-dependent. Their “cancellation” is the cancellation of gauge artifacts in a gauge-invariant combination. This is how gauge invariance works in every other context. Only for θ is it declared mysterious.

APPENDIX E: THE VACUUM IDENTIFICATION ACROSS THE STANDARD MODEL

Sector	Energy functional	Minimum	Vacuum identified with minimum?	Dynamical mechanism demanded?
Higgs	V(φ) = μ ^{2}	φ	^{2} + λ	φ
QED	H QED ground state energy	No real particles	Yes	No
QCD (perturbative)	Λ QCD ground state	Confined vacuum	Yes	No
QCD (topological)	E(θ) = E_0 $-$ χ top cos(θ)	θ = 0	No — “problem” declared	Yes — PQ mechanism demanded

The fourth row is the anomaly. The same identification performed in every other row is refused in the topological sector, and the refusal is called a problem.

APPENDIX F: SERIES PUBLICATION RECORD

Paper	Registry	Key Result
MATH-1	[HOWL-MATH-1-2026]	$\beta = \pi/4$; $Q = F \cdot \beta \cdot d^2 \cdot Z$ across nine domains
MATH-2	[HOWL-MATH-2-2026]	17 transcendentals as integer pairs at 100 digits
MATH-3	[HOWL-MATH-3-2026]	Extended basis: elliptic integrals, Borwein $\zeta(5)$, hierarchy
PHYS-1	[HOWL-PHYS-1-2026]	Mass is inertia; soliton boundaries; three anomaly correlations
PHYS-2	[HOWL-PHYS-2-2026]	Couplings run; transformation law is fundamental
PHYS-3	[HOWL-PHYS-3-2026]	G never measured outside Earth's Hill sphere
PHYS-4	[HOWL-PHYS-4-2026]	Boundary test program; classification; kill switch
PHYS-5	[HOWL-PHYS-5-2026]	α EM running in integer arithmetic; 0.02 ppm
PHYS-6	[HOWL-PHYS-6-2026]	Confinement boundary two-face structure; four SM observables
PHYS-7	**[HOWL-PHYS-7-2026]**	**θ QCD = 0; the strong CP problem dissolves**

END HOWL-PHYS-7-2026

Registry: [HOWL-PHYS-7-2026]

Status: Complete

Domain: Foundational Physics / QCD / Measurement Theory

Central Argument: The strong CP problem is generated by refusing to identify the QCD vacuum with the energy minimum — the same identification performed without comment for every other vacuum in the Standard Model; $\theta\_QCD = 0$ is the ground state of an integer-topological system and requires no mechanism beyond the energy functional

Method: Logical analysis of the problem's premises; identification of the double standard in vacuum identification; examination and demolition of six supporting pillars; two-step response to the basis decomposition objection separating the fine-tuning framing (killed by gauge dependence) from the value question (answered by ground state)

Key Findings: The basis decomposition involves gauge-dependent quantities whose "cancellation" is standard gauge invariance; naturalness has zero confirmed predictions;

CP symmetry protects θ = 0; the institution applies the vacuum = minimum identification everywhere except to θ, and the refusal to apply it is the problem, not the value

Does Not Claim: The axion does not exist; the PQ mechanism is wrong; any new physics

Falsification: Five specific criteria including detection of θ ≠ 0 and identification of observable depending on θ_bare alone

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APPENDIX G: THE COMPLETE VACUUM IDENTIFICATION TABLE

Every vacuum state in the Standard Model, showing the energy functional, the minimum, and whether the institution identifies the vacuum with the minimum or refuses to.

Sector	Energy Functional	Variable	Minimum Location	Vacuum = Minimum?	Problem Declared?	Mechanism Demanded?	Years Spent on “Problem”
Higgs (electroweak)	V(φ) = μ^2	φ	^{2} + λ	φ	^{4}, $\mu^2 <$ 0		φ
QED	H QED = Σ ℏω(k)(a†a + 1/2)	Photon number	n = 0 for all modes	Yes — without comment	No	No	0
QCD (perturbative vacuum)	QCD Hamiltonian ground state	Field configuration	Confined vacuum, Λ QCD ≈ 300 MeV	Yes — without comment	No	No	0
QCD (chiral condensate)	V eff(⟨ q̄q ⟩)	⟨ q̄q ⟩	⟨ q̄q ⟩ ≈ −(250 MeV)^3	Yes — without comment	No	No	0
QCD (topological, θ)	E(θ) = E_0 − χ top·cos(θ)	θ	θ = 0	**No — refused**	**Yes — 50 years**	**Yes — PQ mechanism**	**~50**
Superconducting (BCS)	F(Δ) = F n − N(0)Δ^2/2 + …	Gap Δ	Δ = Δ BCS	Yes — without comment	No	No	0

Sector	Energy Functional	Variable	Minimum Location	Vacuum = Minimum?	Problem Declared?	Mechanism Demanded?	Years Spent on “Problem”
Ferromagnet	$F(M) = aM^2 + bM^4 - H \cdot M$	Magnetization M	M = M sat (at T < T c)	Yes — without comment	No	No	0
Superfluid He-4	F(ψ) below T λ	Condensate ψ		ψ	= √(n 0)	Yes — without comment	No

The pattern: Eight vacuum identifications. Seven are performed without comment. One is refused and declared a problem. The one that is refused is the one where the minimum is at the simplest possible value (zero), protected by a symmetry (CP), confirmed by every measurement, and derivable from the energy functional. The seven that are accepted include values with no derivation, no symmetry protection, and no structural reason.

APPENDIX H: THE TOPOLOGICAL SECTOR STRUCTURE — COMPLETE

The mathematical structure that labels QCD vacua by integers.

Property	Value	Source	Status
Gauge group	SU(3)	Standard Model	Established
Relevant homotopy group	$\pi_3(SU(3))$	Algebraic topology	Mathematical theorem
$\pi_3(SU(3))$ =	$\mathbb{Z}$ (the integers)	Bott periodicity	Mathematical theorem — cannot be violated
Sector label	$n \in \mathbb{Z}$	Winding number = Pontryagin index	Integer by theorem
Instanton charge	$\nu = (1/32\pi^2) \int Tr(G \cdot \tilde{G})\, d^4x$	Topological invariant	Integer for any smooth gauge field
Tunneling between sectors	Via instantons	’t Hooft 1976	Established
Instanton action	$S = 8\pi^2 c_2/g^2 = 256R_4\, c_2/g^2$ (MATH-5)	Gauge theory	Established; R_4 decomposition from MATH-5

Property	Value	Source	Status
θ-vacuum definition		θ ⟩ = Σ n e^(inθ)	n ⟩
Energy functional	$E(\theta) = E_0 - \chi$ top·cos(θ)	Dilute instanton gas / lattice QCD	Established
Topological susceptibility	χ top = $\partial^2 E/\partial\theta^2$	{θ=0} > 0	Lattice QCD; Witten-Veneziano
χ top value	≈ (180 MeV)^4 in pure gauge; ≈ (75 MeV)^4 with quarks	Lattice simulations	Computed to ~5% precision
Ground state	θ = 0 (unique minimum of −cos(θ) on [0, 2 π))	Calculus	Mathematical fact

Every element is from the institution's own mathematics and confirmed by the institution's own computations. The ground state θ = 0 follows from the structure without additional input.

APPENDIX I: THE ENERGY FUNCTIONAL — WHY THE MINIMUM IS AT θ = 0

A complete derivation of E(θ) from its components, showing that every ingredient is established.

Step	Result	Source	Integer Content
1. QCD partition function	Z(θ) = Σ n e^(inθ) Z n	Definition of θ-vacuum	Sum over integer sectors
2. Z n from path integral	Z n = ∫ { ν =n} DA Dq Dq̄ e^{−S}	QCD path integral restricted to sector n	ν = n is integer
3. Free energy	$F(\theta) = -(1/V_4)$ ln Z(θ)	Thermodynamic relation	—
4. Dilute instanton approximation	Z(θ) ≈ exp[V_4 · 2K cos(θ)]	Sum of single-instanton contributions	K = instanton density
5. Energy density	$E(\theta) = E_0 - \chi$ top cos(θ)	χ top = 2K at leading order	—
6. dE/dθ = χ top sin(θ)	Zero at θ = 0 and θ = π	Calculus	—

Step	Result	Source	Integer Content
7. $d^2E/d\theta^2 = \chi$ top $\cos(\theta)$	Positive at $\theta = 0$ (minimum), negative at $\theta = \pi$ (maximum)	Calculus	—
8. χ top > 0	Confirmed by lattice QCD and Witten-Veneziano	Multiple independent calculations	—
9. Therefore	$\theta = 0$ is the unique global minimum on $[0, 2\pi)$	Steps 6-8	—

Lattice confirmation of χ_top > 0:

Lattice Group	Year	Method	χ top^(1/4) (MeV)	χ top > 0?
ETMC	2015	N f = 2+1+1 dynamical fermions	73 ± 4	Yes
Budapest-Wuppertal	2016	N f = 2+1 dynamical fermions	75 ± 3	Yes
MILC	2018	N f = 2+1 staggered fermions	76 ± 5	Yes
Bonati et al.	2015	N f = 2+1 Wilson fermions	74 ± 4	Yes
Pure gauge (quenched)	Various	No dynamical quarks	~180	Yes

Every lattice calculation ever performed confirms χ_top > 0. The minimum is at θ = 0 in every calculation. No exception.

APPENDIX J: THE GAUGE DEPENDENCE — EXPLICIT DEMONSTRATION

Under a chiral rotation q_f → e^(iα_f γ_5) q_f for a single quark flavor f with mass m_f:

Before Rotation	After Rotation	Change
θ bare	θ bare + 2α f	+2α f
arg(m f)	arg(m f) − 2α f	−2α f
θ phys = θ bare + Σ f arg(m f)	θ bare + 2α f + Σ {f'≠f} arg(m {f'}) + arg(m f) − 2α f = θ phys	0 (invariant)
Neutron EDM d n ∝ θ phys	Unchanged	0
η' mass (via χ top)	Unchanged	0
Any CP-violating QCD observable	Unchanged	0

The demonstration for the full quark mass matrix:

Quantity	Expression	Under q f → e^(iα f γ_5) q f	Physical?
θ bare	Coefficient of G·G̃/(32 π ^{2})	θ bare → θ bare + 2 Σ f α f	No — gauge-dependent
det M q	Product of quark masses (complex)	det M q → det M q · e^{−2i Σ f α f}	No — gauge-dependent
arg(det M q)	Phase of quark mass determinant	arg(det M q) → arg(det M q) − 2 Σ f α f	No — gauge-dependent
θ phys = θ bare + arg(det M q)	Physical vacuum angle	Invariant	Yes
Jarlskog invariant J	Im[V us V cb V* ub V* cs] × Δm^2 product	Invariant under all reparametrizations	Yes — weak CP

The "cancellation" between θ_bare and arg(det M_q) is the cancellation of gauge artifacts in a gauge-invariant combination. This is not fine-tuning. This is gauge invariance operating normally. The institution recognizes this cancellation as trivial in every other context (e.g., gauge-dependent pieces canceling in gauge-invariant S-matrix elements) and declares it mysterious only for θ.

APPENDIX K: THE NATURALNESS SCORECARD — EXTENDED

Every prediction naturalness has made, with outcomes.

Prediction	Based On	Year Predicted	Observation Window	Observed?	Status
SUSY at LEP ($\sqrt{s} \leq$ 209 GeV)	Naturalness of Higgs mass	1980s	1989-2000	No SUSY found	Failed
SUSY at Tevatron ($\sqrt{s} \leq$ 1.96 TeV)	Same	1990s	1995-2011	No SUSY found	Failed
SUSY at LHC Run 1 ($\sqrt{s}$ = 7-8 TeV)	Same, updated	2000s	2010-2013	No SUSY found	Failed
SUSY at LHC Run 2 ($\sqrt{s}$ = 13 TeV)	Same, further updated	2010s	2015-2018	No SUSY found	Failed
SUSY at LHC Run 3 ($\sqrt{s}$ = 13.6 TeV)	Same, further updated	2020s	2022-present	No SUSY found so far	Ongoing — expected null
Large θ QCD ~ O(1)	θ is a "random" parameter	1976	1957-present		θ
Λ ~ M Pl^4	Vacuum energy is "natural" at Planck scale	1980s	1998-present	$\Lambda \sim (10^{-3}$ eV$)^4$, off by 10^{120}	Failed
New particles stabilizing m H	m H requires protection	1990s	2012-present	m H = 125 GeV, no partners	Failed
Compositeness at TeV	Higgs as pseudo-Goldstone boson	2000s	2010-present	No compositeness signal	Failed
Extra dimensions at TeV	Hierarchy solved by geometry	1990s-2000s	2010-present	No extra dimension signal	Failed
Dark matter as WIMP at ~100 GeV	Thermal relic with weak coupling	1980s	1990-present	No WIMP detected in 40 years	Failed (at original mass range)

Score: 0 confirmed out of 11 tested predictions. Naturalness as a predictive principle has a 0% success rate across four decades and eleven independent tests. Using naturalness to declare θ = 0 problematic is using a principle with zero predictive power to override a measurement.

APPENDIX L: EVERY MEASUREMENT CONSISTENT WITH θ = 0

Observable	Dependence on θ	Measurement	Bound on	θ
Neutron EDM d n	d n ≈ θ × 3.6 × 10^{-16} e·cm		d n	< 1.8 × 10^{-26} e·cm (Abel et al. 2020)
Mercury EDM d Hg	d Hg ∝ θ via nuclear Schiff moment		d Hg	< 7.4 × 10^{-30} e·cm (Graner et al. 2016)
Electron EDM d e	Indirect sensitivity via θ-induced nuclear effects		d e	< 1.1 × 10^{-29} e·cm (ACME 2018)
$\eta \to \pi^+ \pi^-$	CP-violating decay, branching ratio ∝ θ^2	BR < 1.3 × 10^{-5} (KLOE)	< 0.01	Yes
$\eta' \to \pi^+ \pi^-$	CP-violating decay	Not yet observed	Consistent with θ = 0	Yes
ε’/ε (K meson CP violation)	Dominated by CKM phase; θ contribution suppressed	Measured, consistent with CKM-only	θ contribution below sensitivity	Yes
Nuclear Schiff moments	∝ θ via P,T-violating nuclear forces	All consistent with zero	< 10^{-9} – 10^{-10}	Yes
Molecular EDMs (ThO, HfF^+)	Sensitive to electron EDM and nuclear Schiff	All consistent with zero	Weak constraint on θ directly	Yes
Lattice QCD topological charge distribution	$\langle \nu^2 \rangle = \chi_{top} \cdot V_4$ for θ = 0	Matches θ = 0 prediction	Consistent with θ = 0 exactly	Yes

Observable	Dependence on θ	Measurement	Bound on	θ
η' mass via Witten-Veneziano	$m_{\eta'}^2 = 2N f_{\chi\,top}/f_\pi^2$ at θ = 0	$m_{\eta'}$ = 958 MeV, consistent with χ_{top} from lattice	θ = 0 assumed, consistent	Yes

Total measurements: 10+. Consistent with θ = 0: all. Evidence for θ ≠ 0: none. Every measurement ever performed in every channel at every precision level is consistent with θ being exactly zero.

APPENDIX M: THE AXION SEARCH — 40 YEARS OF NULL RESULTS

Experiment	Method	Mass Range Searched	Coupling Range	Result	Years Active
ADMX	Microwave cavity in magnetic field	1.9-3.5 μeV	DFSZ and KSVZ sensitivity	No axion detected	1996-present
ADMX-SLIC	Sidecar cavity	0.18-0.22 μeV	Below KSVZ	No detection	2019-present
HAYSTAC	Tunable cavity	16.96-17.28 μeV	Near DFSZ	No detection	2015-present
ABRACADABRA	Toroidal magnet, broadband	0.31-8.3 neV	Above DFSZ	No detection	2018-present
CASPEr	Nuclear spin precession	10^{-9}-10^{-6} eV (projected)	Various	No detection (early stages)	2019-present
CAST	Solar axion helioscope	$m_a < 0.02$ eV	$g_{a\gamma\gamma} < 6.6 \times 10^{-11}$ GeV^{-1}	No detection	2003-2015
IAXO (planned)	Next-gen helioscope	$m_a < 0.25$ eV	10 × better than CAST	Not yet operational	Planned

Experiment	Method	Mass Range Searched	Coupling Range	Result	Years Active
ALPS II	Light-shining-through-wall	$m_a < 0.1$ meV	$g_{a\gamma\gamma} < 2 \times 10^{-11}$ GeV^{-1}	No detection (early results)	2023-present
Astrophysical bounds	Stellar cooling, SN1987A	Various	Various	No positive signal; bounds set	1987-present
Cosmological bounds	CMB, BBN, structure formation	Various	Various	No positive signal; bounds set	2000s-present

40+ years of dedicated experimental effort. Multiple independent techniques. No detection. The axion remains a viable dark matter candidate and should be searched for on its own merits. But its non-detection after 40 years cannot be cited as evidence that the strong CP problem is real. An unfound solution to a non-problem is not evidence for the problem.

APPENDIX N: THE HYDROGEN ANALOGY — DETAILED

The analogy between the QCD θ-vacuum and the hydrogen atom is not merely illustrative. The mathematical structures are the same.

Feature	Hydrogen Atom	QCD θ-Vacuum
States		$n \rangle$, $n = 1, 2, 3, \ldots$
Energy	$E_n = -13.6/n^2$ eV	$E(\theta) = E_0 - \chi_{top} \cos(\theta)$
Ground state	$n = 1$	$\theta = 0$
Ground state energy	-13.6 eV	$E_0 - \chi_{top}$
Excited states exist?	Yes — infinitely many	Yes — continuously many
Mechanism to reach ground state demanded?	No	Yes (by the institution)
Problem declared?	No	Yes — for 50 years
Symmetry at ground state	Full rotational symmetry (S state)	CP symmetry
Symmetry broken in excited states?	Yes — angular momentum breaks isotropy	Yes — $\theta \neq 0$ breaks CP
Can system be prepared in excited state?	Yes — by absorbing a photon	In principle — by some unknown mechanism that excites the QCD vacuum

Feature	Hydrogen Atom	QCD θ-Vacuum
Does excited state require explanation?	Yes — must identify the excitation	Yes — $\theta \neq 0$ would require identifying the excitation
Does ground state require explanation?	No — it is the minimum energy	No — it is the minimum energy

The analogy is exact in logical structure. The ground state of a quantum system does not require a mechanism. Excited states require a mechanism (an excitation). The institution applies this logic to hydrogen without hesitation. It refuses to apply it to the QCD vacuum. The refusal generates the "problem."

APPENDIX O: THE BURDEN OF PROOF

Claim	Burden On	Evidence Required	Evidence Available
"The universe is in the ground state ($\theta = 0$)"	No burden — default assumption for quantum systems	Energy functional with minimum at $\theta = 0$	$E(\theta) = E_0 - \chi$ top $\cos(\theta)$, χ top > 0 confirmed by lattice
"The universe is NOT in the ground state ($\theta \neq 0$)"	Claimant — must provide excitation mechanism	Evidence of $\theta \neq 0$ (neutron EDM, η decay, etc.)	None — every measurement consistent with $\theta = 0$
"$\theta = 0$ requires a mechanism (PQ, Nelson-Barr, etc.)"	Claimant — must explain why ground state requires mechanism when no other ground state does	Logical argument distinguishing QCD vacuum from all other vacua	None that survives examination (Section IV-X)
"The cancellation θ bare + arg(det M q) = 0 is fine-tuned"	Claimant — must show both pieces are independently physical	Observable depending on θ bare alone	None — θ bare changes under field redefinition (Section VII)
"Naturalness requires $\theta \sim O(1)$"	Naturalness as a principle	Prior confirmed predictions of naturalness	Zero confirmed predictions in 40 years (Appendix K)

In every other domain of physics, the burden of proof is on those who claim the system is NOT in the ground state. Only for θ has the institution inverted the burden, demanding that the ground state justify itself.

APPENDIX P: WHAT THIS PAPER REMOVES FROM THE SM PARAMETER COUNT

Parameter	Before This Paper	After This Paper	Status
θ QCD	Free parameter, 1 of 19	Derived: $\theta = 0$ (ground state of integer-topological system)	Removed from free parameter count
Remaining free parameters	19	18	One fewer parameter requiring measurement

The 18 remaining parameters:

#	Parameter	Value	Derived?
1	g_1 (U(1) coupling)	~0.358 at M Z	No — measured
2	g_2 (SU(2) coupling)	~0.652 at M Z	No — measured
3	g_3 (SU(3) coupling)	~1.221 at M Z	No — measured
4	λ (Higgs quartic)	~0.129	No — measured (from m H)
5	μ^2 (Higgs mass parameter)	~−(88.4 GeV)^2	No — measured
6	y e (electron Yukawa)	~2.94×10^{-6}	No — measured
7	y μ (muon Yukawa)	~6.09×10^{-4}	No — measured
8	y τ (tau Yukawa)	~1.03×10^{-2}	No — measured
9	y u (up Yukawa)	~1.27×10^{-5}	No — measured (poorly)
10	y d (down Yukawa)	~2.93×10^{-5}	No — measured (poorly)
11	y s (strange Yukawa)	~5.50×10^{-4}	No — measured
12	y c (charm Yukawa)	~7.35×10^{-3}	No — measured
13	y b (bottom Yukawa)	~2.42×10^{-2}	No — measured
14	y t (top Yukawa)	~0.996	No — measured
15	θ_{12} (CKM angle)	~13.0°	No — measured
16	θ_{23} (CKM angle)	~2.36°	No — measured
17	θ_{13} (CKM angle)	~0.201°	No — measured
18	δ CKM (CKM phase)	~1.20 rad	No — measured

θ_QCD was the only parameter with a structural reason to take its value. It is now the first to be removed from the free parameter count. The remaining 18 are genuinely free — they require measurement and have no known derivation from deeper principles (except partially $\sin^2\theta$_W from GUT relations).

APPENDIX Q: CONNECTION TO THE INSTANTON DECOMPOSITION (MATH-5)

MATH-5 showed that the instanton action $S = 8\pi^2 c_2/g^2$ decomposes as $S = 256 R_4 \cdot c_2/g^2$, where $R_4 = \pi^2/32$ is the 4-dimensional n-ball remainder. This connects directly to the θ-vacuum structure.

MATH-5 Object	PHYS-7 Role	Connection
$c_2 \in \mathbb{Z}$ (instanton number)	Topological sector label n	c_2 = n is the winding number; integer by $\pi_3(SU(3)) = \mathbb{Z}$
$R_4 = \pi^2/32$	Geometric content of 4D spacetime	The instanton lives in 4D; R_4 is the 4-ball-to-4-cube ratio
$256 = 8 \times 32$	Normalization ensuring $c_2 \in \mathbb{Z}$	8 = topological normalization; 32 = denominator of R_4
$1/g^2$	Coupling impedance	Determines instanton action magnitude
$S = 256 R_4 c_2/g^2$	Instanton action in the θ-vacuum	Enters $E(\theta)$ through the instanton density $K \propto$ e^{−S}
e^{−S}	Instanton tunneling amplitude	Controls χ top and therefore the depth of the $\theta = 0$ minimum

The chain: $\pi_3(SU(3)) = \mathbb{Z}$ → sectors labeled by integer c_2 → instanton action $S = 256 R_4 c_2/g^2$ → tunneling amplitude e^{−S} → topological susceptibility χ_top > 0 → energy $E(\theta) = E_0 - \chi$_top $\cos(\theta)$ → minimum at $\theta = 0$ → ground state.

Every link is either a mathematical theorem (homotopy group), a geometric identity ($R_4 = \pi^2/32$), an established QFT result (instanton action), or a confirmed lattice computation (χ_top > 0). The ground state $\theta = 0$ follows from the chain without additional input.

APPENDIX R: THE CKM PHASE AND STRONG CP — WHY THEY DON'T CONFLICT

The institution objects that weak CP violation (the CKM phase $\delta\_CKM \approx 1.20$ rad) should "feed into" strong CP violation through arg(det M_q). This appendix shows why the objection fails.

Statement	True or False	Explanation
The CKM phase δ CKM produces CP violation in weak processes	True	Confirmed in K, B, D meson systems
The CKM phase enters arg(det M q)	True	arg(det M q) depends on the Yukawa couplings, which determine both masses and CKM angles
arg(det M q) is a physical observable	**False**	Changes under chiral field redefinition
The CKM phase "feeds into" θ phys	**False**	θ phys = θ bare + arg(det M q) is invariant; the CKM contribution to arg(det M q) is exactly cancelled by the corresponding shift in θ bare under the same field redefinition
Strong CP violation requires θ phys ≠ 0	True	Only θ phys enters strong-sector CP observables
θ phys = 0 contradicts δ CKM ≈ 1.20	**False**	Different observables in different sectors; both are gauge-invariant; both can take their measured values simultaneously
The coexistence of θ phys = 0 and δ CKM ≈ 1.20 requires fine-tuning	**False**	θ phys and δ CKM are independent gauge-invariant observables; one being zero does not constrain the other

The key insight: The Jarlskog invariant $J = \mathrm{Im}[V\_us\ V\_cb\ V^*\_ub\ V^*\_cs] \times \Pi(m\_i^2 - m\_j^2)$ is the unique gauge-invariant measure of CP violation in the weak sector. θ_phys is the unique gauge-invariant measure of CP violation in the strong sector. These are independent gauge-invariant quantities. $J \neq 0$ does not require $\theta\_phys \neq 0$. The Standard Model is perfectly consistent with maximal weak CP violation ($J \approx 3 \times 10^{-5}$) and zero strong CP violation (θ_phys = 0). This is what the data shows.

[**HOWL-PHYS-7-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/H
PHYS-7-2026

[**HOWL-PHYS-1-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/H
PHYS-1-2026

[**HOWL-PHYS-2-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/H
PHYS-2-2026

[**HOWL-PHYS-6-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/H
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[**HOWL-MATH-1-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/MATH
MATH-1-2026

[**HOWL-MATH-2-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/MATH
MATH-2-2026

[**HOWL-MATH-3-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/MATH
MATH-3-2026

[**HOWL-PHYS-3-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/H
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[**HOWL-PHYS-4-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/H
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[**HOWL-PHYS-5-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/H
PHYS-5-2026